

SYNAPSE-SC: System Architecture & Compliance Guardrails

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1. Executive Summary

SYNAPSE-SC is an autonomous, multi-agent orchestration framework designed to mitigate global supply chain disruptions in real-time. Addressing Problem Statement 5 (Domain-Specialized AI Agents with Compliance Guardrails), the system replaces human-in-the-loop latency with deterministic AI tool-calling. Powered by CrewAI and the Llama-3.3-70B model, it ensures high-speed risk resolution while strictly adhering to enterprise financial compliance.

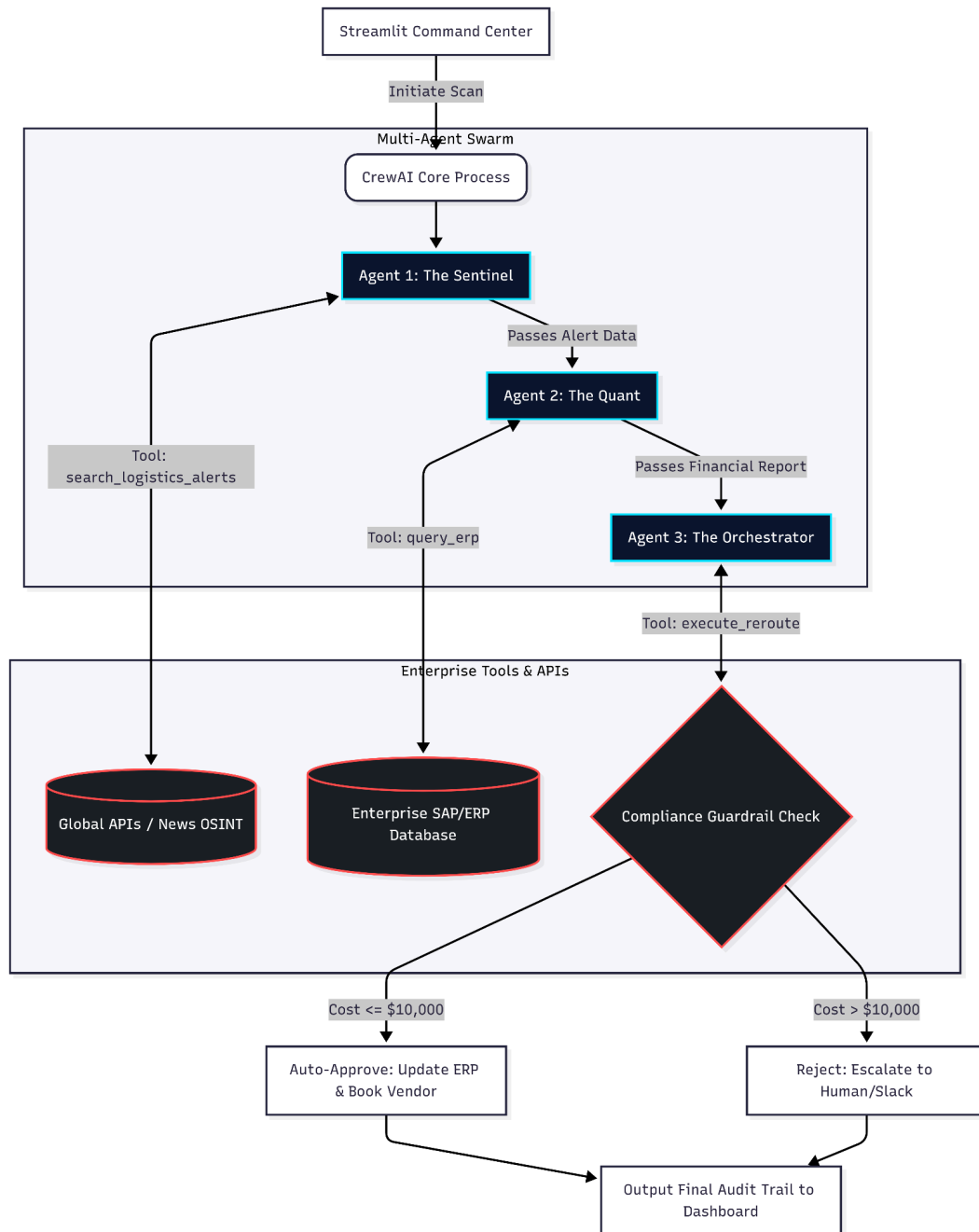
2. Multi-Agent Topology & Sequential Data Flow

The system executes a "Chain of Thought" pipeline utilizing three specialized AI personas:

- **Agent 1: The Sentinel (Signal Detection)**
 - **Function:** Acts as the early-warning radar. It connects to external OSINT and global weather/logistics APIs, filtering noise to output verified disruption alerts (e.g., "Typhoon halted Port of Shenzhen").
- **Agent 2: The Quant (Impact Modeler)**
 - **Function:** Translates external events into internal financial exposure. It cross-references the Sentinel's alert with internal SAP/ERP databases to identify delayed shipments, calculate at-risk inventory value, and project total SLA penalties.
- **Agent 3: The Orchestrator (Decision & Execution)**
 - **Function:** The decisive supply chain commander. It ingests the financial damage report, evaluates available backup logistics vendors, calculates mitigation costs, and executes the rerouting API order.

3. Automated Execution Workflow

1. **Initiation:** User triggers the command center scan.
2. **Ingestion:** Sentinel Agent fetches real-time API data.
3. **Analysis:** Quant Agent maps data to enterprise ERP.
4. **Decision:** Orchestrator Agent selects optimal backup vendor.
5. **Validation:** Hard-coded tool intercepts the decision to verify guardrails.
6. **Execution:** If approved, ERP is updated; if rejected, human fallback is triggered.



4. Hard-Coded Compliance & Guardrails (Risk Mitigation)

GenAI is unreliable in enterprise environments without strict programmatic boundaries. SYNAPSE-SC utilizes deterministic, code-level constraints rather than relying solely on LLM prompt instructions:

- **Financial Spend Guardrail:** The `execute_reroute` Python tool contains hard-coded financial logic. If the AI orchestrator attempts to book a vendor exceeding the \$10,000 limit, the tool overrides the AI, rejects the API call, and flags the transaction for human review on Slack/Teams.
- **Vendor Whitelisting:** The execution layer is restricted to interacting only with a pre-approved array of secondary logistics providers, physically preventing rogue external API calls.
- **Immutable Auditability:** Every reasoning step, API invocation, and calculated parameter generated by the agent swarm is logged sequentially. This creates a transparent, compliance-ready audit trail for enterprise review.

5. Technology Stack

- **Frontend:** Streamlit (Python) for real-time command-center visualization.
- **Orchestration:** CrewAI for sequential task delegation and multi-agent memory.
- **Core LLM Engine:** Llama-3.3-70B (via Groq + LiteLLM bridging) for near-zero latency reasoning.
- **Integration Layer:** Native CrewAI tool wrappers simulating external API and internal ERP database access.